

Sri Lanka Institute of Information Technology



<Exercise 5>

<IT25102533>

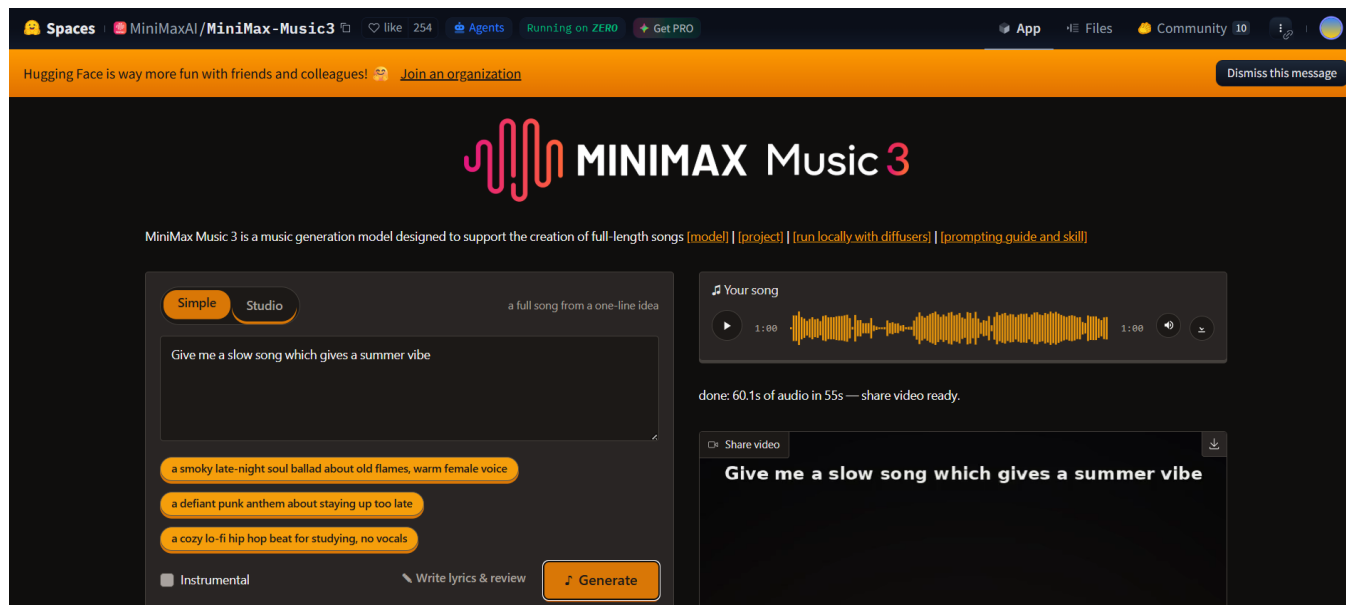
<I.K Divesh>

Artificial Intelligence and Machine Learning

IT2011

B.Sc. (Hons) in Information Technology

Hugging Face Spaces



Google Colab

9.

```
from transformers import pipeline
summarizer = pipeline("summarization")
summary = summarizer("""Hugging Face is a pioneering open-source AI platform
that empowers developers, researchers, and organizations to easily access and
deploy cutting-edge machine learning models for natural language processing, computer vision,
and beyond—democratizing AI and accelerating innovation through a collaborative ecosystem of tools,
datasets, and community-driven contributions.""", max_length=20)
print(summary)
```

```
... No model was supplied, defaulted to sshleifer/distilbart-cnn-12-6 and revision a4f8f3e (https://huggingf
Using a pipeline without specifying a model name and revision in production is not recommended.
config.json: 1.80k/? [00:00<00:00, 38.3kB/s]
pytorch_model.bin: 100% 1.22G/1.22G [00:16<00:00, 97.8MB/s]
tokenizer_config.json: 100% 26.0/26.0 [00:00<00:00, 1.22kB/s]
vocab.json: 899k/? [00:00<00:00, 25.7MB/s]
merges.txt: 456k/? [00:00<00:00, 21.9MB/s]
Your min_length=56 must be inferior than your max_length=20.
/usr/local/lib/python3.13/dist-packages/transformers/generation/utils.py:1298: UserWarning: Unfeasible l
warnings.warn(
[{'summary_text': ' Hugging Face is a pioneering open-source AI platform that empowers developers, rese
```

11.

```
from transformers import pipeline

# Use a known summarization model
summarizer = pipeline("summarization", model="EbanLee/kobart-summary-v3")

# Input text to summarize
text = """Hugging Face is a pioneering open-source AI platform
that empowers developers, researchers, and organizations to easily access and
deploy cutting-edge machine learning models for natural language processing, computer vision,
and beyond—democratizing AI and accelerating innovation through a collaborative ecosystem of tools,
datasets, and community-driven contributions."""

# Generate summary
summary = summarizer(text, max_length=20, min_length=5, do_sample=False)
print(summary)
```

... You passed along `num\_labels=3` with an incompatible id to label map: {'0': 'NEGATIVE', '1': 'POSITIVE'}. The number of labels will be overwritten to 2.
You passed along `num\_labels=3` with an incompatible id to label map: {'0': 'NEGATIVE', '1': 'POSITIVE'}. The number of labels will be overwritten to 2.
[{'summary_text': 'Hugging Face is a pioneering op'}]

13. Sentiment Analysis

```
from transformers import pipeline

# Load a specific sentiment analysis model from Hugging Face
classifier = pipeline("text-classification", model="distilbert-base-uncased-finetuned-sst-2-english")

# Testing the model with a custom sentence
result = classifier("I absolutely love learning about Artificial Intelligence and passing my university labs!")

print(result)
```

... config.json: 100%  629/629 [00:00<00:00, 62.4kB/s]
model.safetensors: 100%  268M/268M [00:01<00:00, 236MB/s]
tokenizer_config.json: 100%  48.0/48.0 [00:00<00:00, 4.36kB/s]
vocab.txt:  232k/? [00:00<00:00, 6.32MB/s]
[{'label': 'POSITIVE', 'score': 0.9995057582855225}]